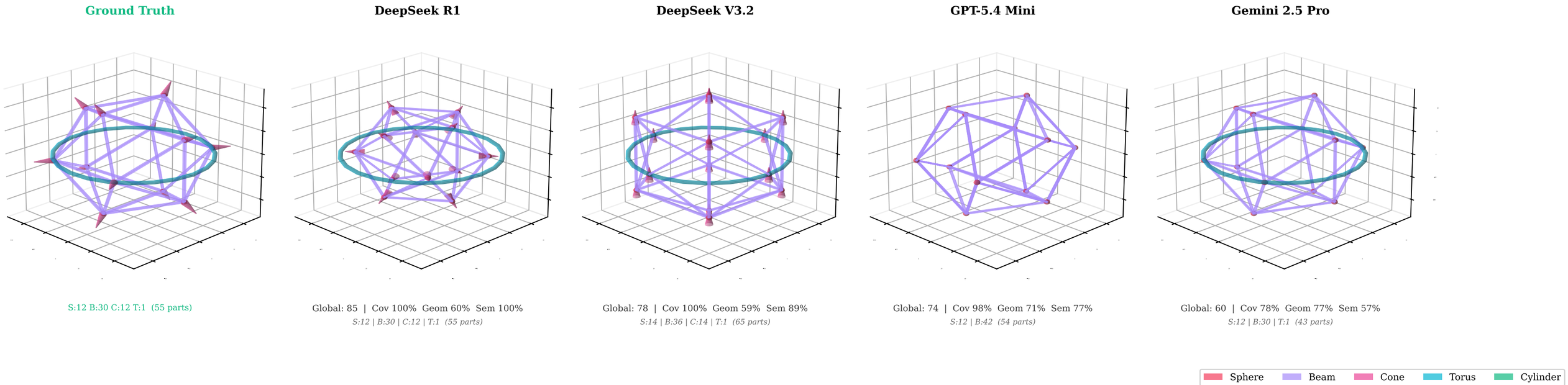

Figure 21: Radiolarian Skeleton (L1, 55 shapes) — qualitative comparison across four models.
Ground truth: 12 vertex spheres (red) + 30 edge beams (purple) + 12 radial spine cones (pink) + 1 equatorial torus (cyan).
Key failure: Only DeepSeek R1 produces cones. Most models omit all 12 radial spines, losing ~22% of shapes and critical structural identity.